

KANU VICTOR IKENNA

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PROFILE

Biochemistry-trained computational biomedical researcher working at the intersection of AI-driven drug discovery, blockchain-secured provenance, and digital twin modeling. My current research develops verifiable biomedical workflows for protein analysis, virtual screening, biomaterial provenance, and stem-cell differentiation forecasting. For doctoral research, I aim to build trustworthy AI digital twins for stem-cell-derived Parkinson's disease models, integrating multi-omics, perturbation prediction, and auditable data/model provenance.

Research Interests: Structure-based virtual screening, molecular activity prediction, protein–ligand modelling, machine learning for drug discovery, and computational biochemistry.

EDUCATION

Kumoh National Institute of Technology, Gumi, Gyeongsangbuk, South Korea

- Master of Engineering, ICT Convergence Engineering | 4.44/4.5 Sept. 2024 – Aug. 2026

University of Lagos, Akoka, Lagos, Nigeria

- Bachelor of Science, Biochemistry | CGPA: 3.90/5.0 Nov. 2015 – Jan. 2020

Yaba College of Technology, Yaba, Lagos, Nigeria

- National Diploma, Food Technology | CGPA: 3.23/4.0 Jul. 2013 – Nov. 2015

The Bridge Program, Lagos, Nigeria

- Entrepreneurial Leadership July – Sept 2016

ACADEMIC AWARDS, HONORS, AND ACHIEVEMENTS

Outstanding Academic Papers Award , presented by Kumoh National Inst. of Technology	Aug. 2026
3rd Place, NSL Winter Intensive Papers , presented by NSLab	Apr. 2026
Excellent Paper Award, ICAIIC 2026, Tokyo Japan , presented by the ICAIIC TPC	Feb. 2026
Grand Prize Winner, Gumi Industrial Complex Paper Contest , presented by KICS	Dec. 2024
Best Essay Award , presented by Gyeongbuk Provincial College	Aug. 2024
K-GKS Scholarship , a full-ride graduate scholarship provided by the Korean Government	July. 2024
The Lead Student Innovators Award , presented by the Vice-Chancellor of the University of Lagos	Sept. 2019
Best Speaker Award, Biochemistry Inter-Level Debate , presented by Biochemistry Department	Oct. 2018
The Most Viable Idea Prize ; presented by The Bridge Program.	Sept. 2016
Best Group Presentation , presented by the Entrepreneurial Development Program, Yabatech.	Aug 2015

RESEARCH EXPERIENCE, PUBLICATIONS AND PRESENTATIONS

Manuscript published/under review.

1. **Kanu, V.I.**, Nwadiugwu, P.W., Kim, D.S. Proteomics-Integrated Digital Twin State Space for Dopaminergic Stem Cell Differentiation, 2026. (Under review). *Computer Methods and Programs in Biomedicine*
2. **Kanu, V.I.**, Ajakwe, U.I., Isong, J.A., Ajakwe, S.O., Kim, D.S. A Leakage-Safe Benchmark for Single-Cell Differentiation Forecasting, 2026. (Under review). *Scientific Reports*
3. **Kanu, V.I.**, Ajakwe, S.O., Kim, D.S. PureProtX: A PoA²-Secured Deterministic Edge Inference Framework for Verifiable IoT-Distributed Scientific Computation, 2026. (Major revision). *IEEE Internet of Things*
4. **Kanu, V.I.**, Isong, J.A., Ajakwe, S.O., Kim, D.S. BioPassport: Edge-Attested Biomaterial Provenance for IoT-Integrated Laboratories, 2026. (Minor revision). *IEEE Internet of Things*

5. [Kanu, V.I., Ajakwe, S.O., Lee, J.M., Kim, D.S.](#) Deterministic Protein Structure and Binding Site Analysis through Blockchain-Integrated Workflow Verification, 2025. *ICT Express*
6. [Kanu, V.I., Isong, J., Ajakwe, S.O., Kim, D.S.](#) ProtChain: Blockchain-Secured IoT Data Provenance for Biomedical and Scientific Workflows, 2025. *Peer J Computer Science*
7. Amadi, S.C., **Kanu, V.I.**, Ajakwe, S.O., Kim, D.S, Taesoon, J. CONQUEST: A Federated Context-Aware Quantum-Inspired Framework for Robust Cyberworm Detection in Mission-Critical IoT Networks, 2026. (Under review)
8. Amadi, S.C., **Kanu, V.I.**, Ajakwe, S.O., Kim, D.S, Taesoon, J. OrBIT: Orchestrated Blockchain Integrated Twins for privacy-aware Edge Intelligence and Transparent Lifecycle, 2026. (Under review). *IEEE Internet of Things*
7. Isong, J.A., **Kanu, V.I.**, Ajakwe, S.O., Kim, D.S. FedEpi: A Robust Physics-Informed Federated Framework for Secure Multi-Source IoT Intelligence, 2026. (Under review)
9. [Ajakwe, I.U., Kanu, V.I., Ajakwe, S.O., Kim, D.S.](#) PredBlock: A Physics-Informed AI and Blockchain Framework for Integrity Verification in CO₂ CCS Pipelines Optimization. 2026 *International Journal of Greenhouse Gas Control*, 104696
10. [Kanu, V.I., Ajakwe, I.U., Ajakwe, S.O., Kim, D.S.](#) Offline-Capable AI-Blockchain Architecture for Biochemical Threat Detection in Mission-Critical MANET Environments, 2025. *IEEE Internet of Things*
11. [Ajakwe, I.U., Kanu, V.I., Ajakwe, S.O., Kim, D.S.](#) eBCTC: Energy-Efficient Hybrid Blockchain Architecture for Smart and Secured K-ETS, 2025. *Cleaner Engineering and Technology*, 101084
12. [Ihejirika, M., Osaretin, A. T. E., Imaga, N. A., Kanu, V. I., & Edmunds, P. O.](#) Phytochemical and antioxidant profiling of aqueous and ethanol extracts of Curcuma longa and Cinnamomum zeylanicum, 2022. *International Journal of Advance Biochemical Research*.

Conference Proceedings

1. [Kanu, V.I., Ajakwe, U.I., Isong, J.A., Ajakwe, S.O., Kim, D.S.](#) Transformer-Based Digital Twin for Stem Cell Differentiation Forecasting with Mechanistic ODE Benchmarking. In *17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, July 7-10, 2026*
2. [Kanu, V.I., Ajakwe, S.O., Kim, D.S.](#) BioPassport: A Policy-Enforced Blockchain-Credential Architecture for Tamper-Evident Biomaterial Provenance. In *8th edition of the IEEE International Conference on Blockchain and Cryptocurrency in Brisbane, Australia, 1-5 June 2026*.
3. [Kanu, V.I., Ajakwe, S.O., Kim, D.S.](#) VeraComm: Verifiable Communication Protocol for AI-Based Drug Screening using Blockchain. In *8th International Conference on AI in Information and Communication, Tokyo, Japan, Feb 24-27, 2026*
4. [Kanu, V.I., Ajakwe, S.O., Kim, D.S.](#) PureProt: An AI-Blockchain Enabled Virtual Screening Platform for Drug Discovery. In *16th IEEE International Conference on ICT Conference, ICTC, Jeju Island, South Korea, Oct 2025*.
5. **Kanu, V.I.**, Ajakwe, I.U., Ajakwe, S.O., Kim, D.S.. Tamper-Resistant AI-Blockchain-Driven Biochemical Threat Detection in Disconnected Tactical MANETs. In *Young Scholars Workshop (YSW), IEEE Military Communications Conference (Accepted and to be presented) in October 6-10, 2025*.
6. [Kanu, V.I., Isong, J., Ajakwe, S.O., Kim, D.S.](#) Blockchain-Enabled Framework for Efficient and Interoperable Proteomic Data Management. In *International Conference on Ubiquitous and Future Networks (ICUFN), Lisbon, Portugal, July 6-11, 2025 [Online Presentation]*
7. **Kanu, V.I.**, Ajakwe, S.O., Kim, D.S. Blockchain-Verified Deterministic Framework for Protein Structure and Binding Site Analysis in Drug Discovery Distributed System. In *Korean Institute of Communications and Information Sciences (KICS) Summer Conference 2025, Jeju Island, South Korea, June 18-20, 2025*
8. [Kanu, V.I., Isong, J., Ajakwe, S.O., Kim, D.S.](#) Microservices Architecture for Blockchain-Based Proteomic Data Storage System Integrated with PDB and UniProt. In *Korean Institute of Communications and Information Sciences (KICS) Winter Conference 2025, Pyeongchang, South Korea February 5-7, 2025*

9. Nnadiakwe, C.A., **Kanu, V.I.**, Akor, G., Kim, D.S. (2025). Blockchain-enabled Secured Light Weight Data Management in Smart Factories. In Korean Institute of Communications and Information Sciences (KICS) *Winter Conference 2025*, February 5-7, 2025
10. **Kanu, V.I.**, Ajakwe, U.I., Ajakwe, S.O., Lee, J.M., Kim, D.S. Anthropomorphic and Accountable AI-Driven Power Generation and Distribution System. In *KICS Fall Conference 2025*, November 20-22, 2024
11. [Kanu, V.I.](#), [Ajakwe, S.O.](#), [Lee, J.M.](#), [Kim, D.S.](#) Blockchain Technology in Protein Folding: Enhancing Data Sharing and Collaboration. In *15th IEEE International Conference on ICT Conference, ICTC*, October 16-18, 2024. doi: 10.1109/ICTC62082.2024.10826960

Seminar Presentations

1. R Tapwal, S Misra, SK Pal. UtilityChain: Dynamic Resource Allocation for Mining and Servicing in Blockchain System. In *IEEE Transactions on Services Computing*, 2023. **Presented 2024. 09.11**
2. AM Yakubu, YPP Chen. A Blockchain-based Application for Genomic Access and Variant Discovery using Smart Contracts and Homomorphic Encryption. *Published in Elsevier Future Generation Computer Systems*, 2022. **Presented 2025. 03.20**
3. U Demirbaga, GS Aujla. MapChain: A Blockchain-Based Verifiable Healthcare Service Management in IoT-Based Big Data Ecosystem. *Published in IEEE Transactions On Network And Service Management*, Vol. 19, No. 4, December 2022. **Presented 2025. 07. 10**
4. R Wang, C Xu, X Zhang. Toward Materials Genome Big-Data: A Blockchain-Based Secure Storage and Efficient Retrieval Method. *Published in IEEE Transactions on Parallel and Distributed Systems*, 2024. **Presented 2025.09. 04**
5. XX Du, MY Liu, YH Sun. Segmentation, Detection, and Tracking of Stem Cell Image by Digital Twins and Lightweight Deep Learning. *Published in Computational Intelligence and Neuroscience*, 2022. **Presented 2026.03. 02**

Book Chapter

1. [Okafor, C.I.](#), [Kanu, V.I.](#), [Akor, C.G.](#), [Lee, J.M.](#), [Kim, D.S.](#) Autonomous Intelligent Vehicles: An Overview. In *Blockchain For Security of Autonomous Intelligent Vehicles*, to be published in *Blockchain for Security of Autonomous Intelligent Vehicles. Studies in Big Data*, vol 194. Springer, Cham.

Selected Relevant Projects/Presentations

1. **Blockchain-Enabled Trust Infrastructure for Verifiable Biomolecular Computational Workflows** May 21, 2026
Master of Engineering Thesis Presentation: This thesis investigates how biomolecular computational workflows (BCWs) can be engineered as verifiable systems that integrate tamper-evident provenance, deterministic output verification, and policy-enforced validity for multi-party exchange.
2. **PureProt / PureProtX** Dec 2025 - Present
AI-enabled virtual screening and protein-focused drug discovery workflow integrating computational modeling and predictive analysis. Worked on target-dependent screening concepts, protein structure/binding-site analysis, and reproducible workflow design.
3. **ProtChain** Sept 2024 - Jan 2026
Blockchain-secured framework for biomedical/scientific workflow provenance, integrating protein data resources such as PDB and UniProt; relevant to reproducible computational biology pipelines.
4. **HackBio Bioinformatics Internship Project** Sept 2021
Investigated protein targets of Mycobacterium tuberculosis and explored potential drugs using molecular docking and bioinformatics tools.

Lab Projects

Tech Lead, <i>HiTex</i> , Secure collaborative LaTeX writing tool built on blockchain (Purechain)	Jan 2026 -present
Tech Lead, <i>ChainPilot</i> , a blockchain abstraction platform that would facilitate the adoption of the Lab's blockchain network - Purechain.	Aug-Nov. 2025
Technical Paper Lead, <i>PureCertificate</i> , certificate protection and tampering verification through blockchain	Mar. 2024 - Dec 2025

Other Paper Submissions

Victor Ikenna Kanu. A proposed solution that conveys how the nature of work is changing and what the Nigerian Government can do to prepare and support its workforce. A Digital Artifact submitted to the World Bank Group's MOOC on EDx <https://sites.google.com/view/da-futureofwork/home>

PROFESSIONAL AFFILIATIONS

- **Student Member**, Institute of Electrical and Electronics Engineers (IEEE) - No: 102105174
- **Member**, IEEE Engineering Medicine and Biology Society (IEEE EMBS)
- **Member**, Korean Institute of Communications and Information Sciences (KICS)
- **Member**, Nigerian Institute of Food Science and Technology (NIFTS)
- **Member**, Nigerian Society of Biochemistry Students (NSBS)

TEACHING/ADMINISTRATIVE EXPERIENCE

The Bridge Program - Entrepreneurial Leadership Facilitator	Aug 2021 & Aug 2022
● Facilitated sessions in Empathy & Feedback, Identity & Mental Models, Proactivity, and Leading Change to the assigned class of students selected from 102 universities in Nigeria in their journey to be program fellows. ● Taught classes on Negotiations, Context Awareness & Needs Identification, and Leading Self to the assigned class of students selected from 102 universities in Nigeria in their journey to be program fellows.	
Breakthrough Tutorial - Biology and Chemistry Tutor	June 2013 – Mar 2022
● Prepared over 100 students in Biology and Chemistry for national secondary school and university entrance examinations; the current passing rate is at 90%.	
Elizade University ELP - Administrative Assistant	June 2013 – Mar 2022
● Supported the program facilitator in preparing teaching and training documents for the ELP lecturers of Elizade University, Ondo State.	

WORK EXPERIENCE

Network Systems Lab, Kumoh National Institute of Technology – Graduate Researcher	Sept 2024 - Present
● Conduct research on blockchain-enabled biomedical/scientific workflow provenance, AI-assisted drug discovery systems, and verifiable computational pipelines. ● Develop and evaluate frameworks integrating protein data resources, IoT-enabled laboratory data, smart contracts, and distributed ledger technologies. ● Contribute to manuscript development, experimental design, system architecture, benchmarking, and technical documentation for journal and conference publications. ● Collaborate with NSLab researchers on projects involving PureChain, ProtChain, PureProt/PureProtX, BioPassport, and secure edge intelligence.	

IXO Gadgets – Product Researcher

Feb 2023– Jun 2024

- Conducted thorough market research, including taking surveys, reading and perusing tech gadget releases and doing desktop research, which then developed into an ongoing holistic report compendium of about 50 products.
- Collaborated and strategised with the company founders and the marketing team on new trends and demographics to target with with Ads and marketing. This synergy has resulted in a 30% sales jump in 2 months.
- Planned and assisted with new product releases, identified new market opportunities and relayed this information to the Manager.

Future Females Invest – Programme and Business Development Associate

Feb 2021– Aug 2022

- Supported business development by writing proposals and preparing presentation slides.
- Researched over 1000 targeted campaigns for partnerships, business development, thought leadership, and speaker confirmation.
- Developed and curated new programs that best suited the objectives of the business while researching over 600 organisations and individuals to find new opportunities, which resulted in a 60% increase in new partnerships and a 20% increase in revenue.

Seven-Up Bottling Company - Quality Control Intern

Oct. – Dec. 2018

- Inspected output samples using industry-appropriate methods, such as comparing standards, measuring dimensions, and examining the functionality.
- Checked that the assembly or production line adheres to standards and procedures and complies with legal requirements.
- Proposed improvements to the production process.

Lagos State Ministry of Health - Nutrition Support Intern

Apr. – Jul. 2015

- Organised and facilitated vaccination and immunization programs for children (0 months to 2 years), resulting in the drop of polio cases to 0.1% that year which eventually led to being certified Polio-free State.
- Advocated for exclusive breastfeeding of children under the age of 2, based on research findings.

LEADERSHIP/VOLUNTEER EXPERIENCE

Program Manager & Selection Lead, Professor Ogundipe Innovative Challenge

Aug. 2019 - Feb. 2020

Program and Project Manager, The Bridge Program (Educate-to-Employ program)

Dec 2017 - Dec. 2018

Strategic Planning Lead, KidsAide Initiative

May - Dec. 2018

SKILLS AND ABILITIES

Languages: English, Yoruba, and Igbo Language: Strong reading, writing, and speaking competencies. Korean and Spanish: Introductory level at reading, speaking and writing.

Software/Coding: Data Cleaning. Molecular Dynamics Simulations. Data Visualization, SQL, Sequence Alignment, Latex, Github, Microsoft Office Suite (Word, Excel, PowerPoint, etc.), Protein engineering experiments, protein structure data workflows.

Soft Skills: Academic Research, Teaching, Project Management, Leadership, and Team Management

COURSES AND CERTIFICATIONS

International English Language Testing System (IELTS)

May 2024 - May 2026

Total Score = 7.5

- Speaking: 7.5
- Listening: 7.5

- Reading: 8.0
- Writing: 6.5

University of California San Diego Bioinformatics Coursera Scholarship (Certification) July 2022 - June 2023

- Investigated DNA replication to determine where the genome replication begins -OriC.
- Determined the sequence of amino acids making up one of the investigated antibiotics using brute force algorithms.
- Applied popular bioinformatics software tools to sequence the genome of a deadly Staphylococcus bacterium that has acquired antibiotic resistance.

Mar 2021 -Feb 2022

Google Data Analytics Professional Certificate Coursera Scholarship (Certification)

- Explored a variety of real-world business scenarios to support an understanding of questioning and decision-making.
- Discovered how to identify different types of bias in data to help ensure data credibility.
- Applied basic SQL functions for cleaning and transforming data

Aug. - Sept. 2021

Hackbio Bioinformatics Virtual Internship Certificate

- Learned and applied basic Python codes in analyzing biodata.
- Investigated the target protein of *Mycobacterium tuberculosis* and the potential and existing drugs using molecular docking and other bioinformatics software tools.
- Over 400 phyto alkaloids that can serve as potential inhibitors to bacterial meningitis-causing pathogens were investigated using *Insillico* analysis.

REFERENCES

Simeon Okechukwu Ajakwe, Ph.D. (Research Mentor)

*Associate Professor, The Smart Computing Dept,
Kyungdong University, Global Campus, South Korea.*

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Charles Pickens, Ph.D. (Mentor)

*Professor, Center for Cognitive and Neurobiological
Approaches to Plasticity (CNAP), Kansas State University*

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Williams Paul Nwadiugwu, Ph.D. (Research Mentor)

*Professor, Bharti School of Engineering,
Laurentian University, ON P3E 2C6, Ontario, Canada.*

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Ridwan Abiodun Lawal, Ph.D. (Undergrad Supervisor)

*Associate Professor, Dept. of Biochemistry
University of Lagos, Akoka-Yaba, Lagos, Nigeria*

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